AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

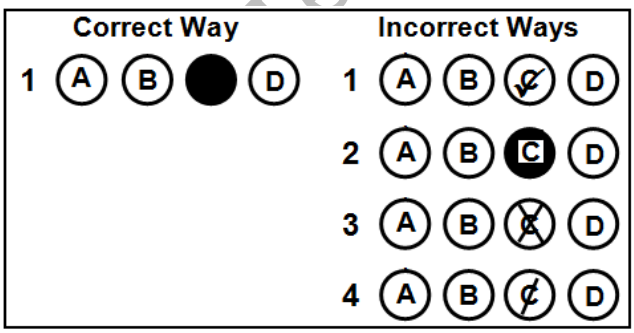
Mathematics Paper I

Time: 45 minutes Marks: 35

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 35 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The function $f(x) = \sin x \cos x$ is a/ an

   I. odd function.
   II. even function.
   III. parametric function.

   A. I only
   B. II only
   C. I and III
   D. II and III

2. The function $f(x) = \frac{2}{x-1}$ has the domain $R - \{1\}$. The range of $f^{-1}(x)$ is

   A. R
   B. $R - \{0\}$
   C. $R - \{1\}$
   D. $R - \{2\}$

3. If $f(x) = x - 1$ and $g(x) = 2x^2$, then $f(g(x))$ is equal to

   A. $2x^3 - 1$
   B. $2x^2 - 1$
   C. $2(x-1)^2$
   D. $2x(x-1)^2$

4. The Tylor series of a function $f(x)$ about a point $x = a$ is defined as follows.

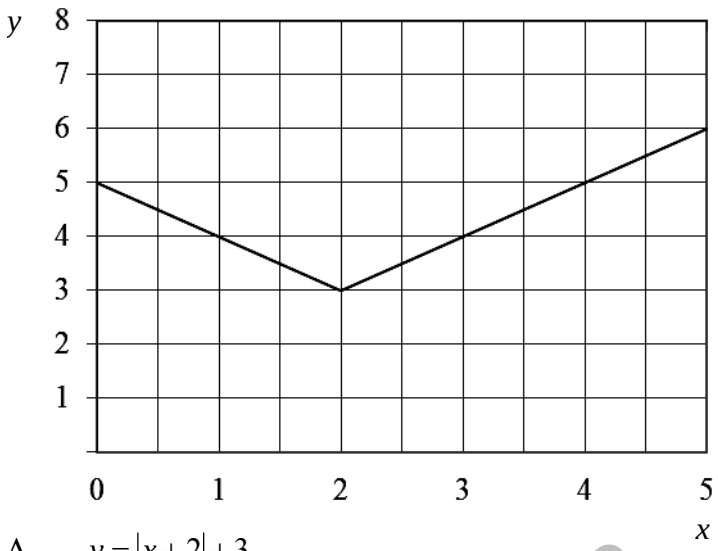
   $$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots$$

   If $f(x) = \ln(x+1)$, defined in the interval containing '3', is expanded as Taylor series at $a = 3$, and $f'(3) = \frac{1}{4}$, then the second term of this series will be

   A. $\frac{(x+3)}{4.2!}$
   B. $\frac{(x-3)}{4.2!}$
   C. $\frac{1}{4}(x+3)$
   D. $\frac{1}{4}(x-3)$

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5. The given figure shows a modulus function that can be represented by the equation



A. $y = |x+2|+3$

B. $y = |x+2|-3$

C. $y = |x-2|+3$

D. $y = |x-2|-3$

6. If $y = \frac{x^2}{3}$, then $\frac{d^3y}{dx^3}$ is equal to

A. 0

B. $\frac{2x}{3}$

C. x

D. $2x$

7. For $y = 3x$, when $x = 1$ and $\delta x = 0.1$, the value of δy is

A. 0.03

B. 0.33

C. 0.3

D. 3

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8. If $p = e^{2x}$ and $q = 3x$, then $\frac{dp}{dq}$ is equal to

A. $\frac{e^{2x}}{6}$

B. $\frac{2e^{2x}}{3}$

C. $\frac{3}{2e^{2x}}$

D. $6e^{2x}$

9. If $f(x) = \ln(a)$, where a is an integer, then the derivative of $f(x)$ with respect to x is

A. 0

B. 1

C. a

D. $\frac{1}{a}$

10. The derivative of $\cos^{-1} x$ with respect to x is

A. $\frac{1}{\sqrt{1+x^2}}$

B. $\frac{1}{\sqrt{1-x^2}}$

C. $-\frac{1}{\sqrt{1+x^2}}$

D. $-\frac{1}{\sqrt{1-x^2}}$

11. $\frac{1}{(2x+3)}$ and $\frac{1}{(x-3)}$ are the partial fractions of which of the following rational expressions?

A. $\frac{3x+6}{(2x+3)(x-3)}$

B. $\frac{-x-6}{(2x+3)(x-3)}$

C. $\frac{3x}{(2x+3)(x-3)}$

D. $\frac{2}{(2x+3)(x-3)}$

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12. $\int 9dx$ is equal to

A. $9c$
B. $9 + c$
C. $\frac{x}{9} + c$
D. $9x + c$

13. $\int_5^5 dx$ is equal to

A. 0
B. 1
C. 10
D. 25

14. If $\int_2^3 f(x)dx = 7$, then $\int_3^2 f(x)dx$ is equal to

A. -7
B. -1
C. 1
D. 7

15. A ball rolls in a straight line such that t seconds after passing through a fixed point O, its displacement is given by $3t + t^2$ m. The velocity of the ball is given by

A. $t + 2t$ ms^{-1}
B. $3 + 2t$ ms^{-1}
C. $6t^2 + 3t^3$ ms^{-1}
D. $\frac{3}{2}t^2 + \frac{t^3}{3}$ ms^{-1}

16. $\int_1^2 \frac{1}{x^2} dx$ is equal to

A. $-\frac{3}{2}$
B. $-\frac{1}{2}$
C. $\frac{1}{2}$
D. $\frac{3}{2}$

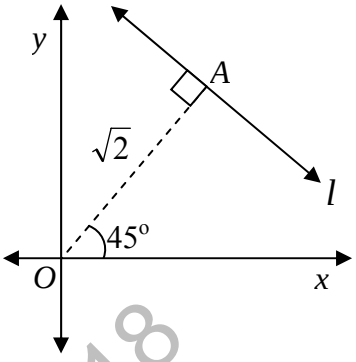
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17. The angle between tangent and normal, at point (0, 10) on the circle $x^2 + y^2 = 100$, is

A. 0°
B. 30°
C. 60°
D. 90°

18. In the given figure, *OA* is perpendicular to line *l*. If the length of *OA* is $\sqrt{2}$ units, then the equation of line *l* is

A. $x + y + 2 = 0$
B. $x + y - 2 = 0$
C. $x - y + 2 = 0$
D. $x - y - 2 = 0$



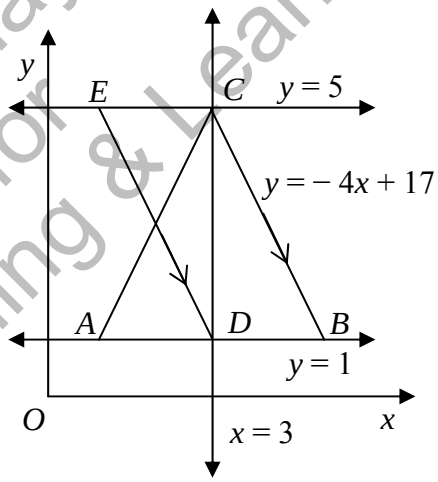
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In the given diagram, triangle *ABC* is an isosceles triangle. Equation of the line *AB* is $y = 1$, *EC* is $y = 5$, and *CB* is $y = -4x + 17$. Line *CB* is parallel to line *ED*.

Using the given information, answer Q.19, Q.20 and Q.21.

19. The altitude of triangle *ABC* is

A. 3 units
B. 4 units
C. 5 units
D. 6 units



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20. The equation of the line *ED* is $y = mx + 13$. The value of *m* is

A. – 4
B. – 1
C. 1
D. 4

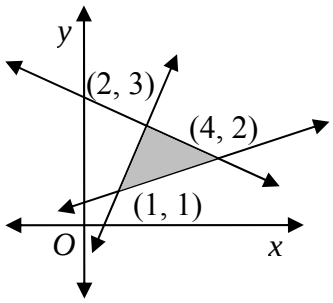
21. The distance of line *CD* from *y*-axis is

A. 1 unit.
B. 3 units.
C. 4 units.
D. 5 units.

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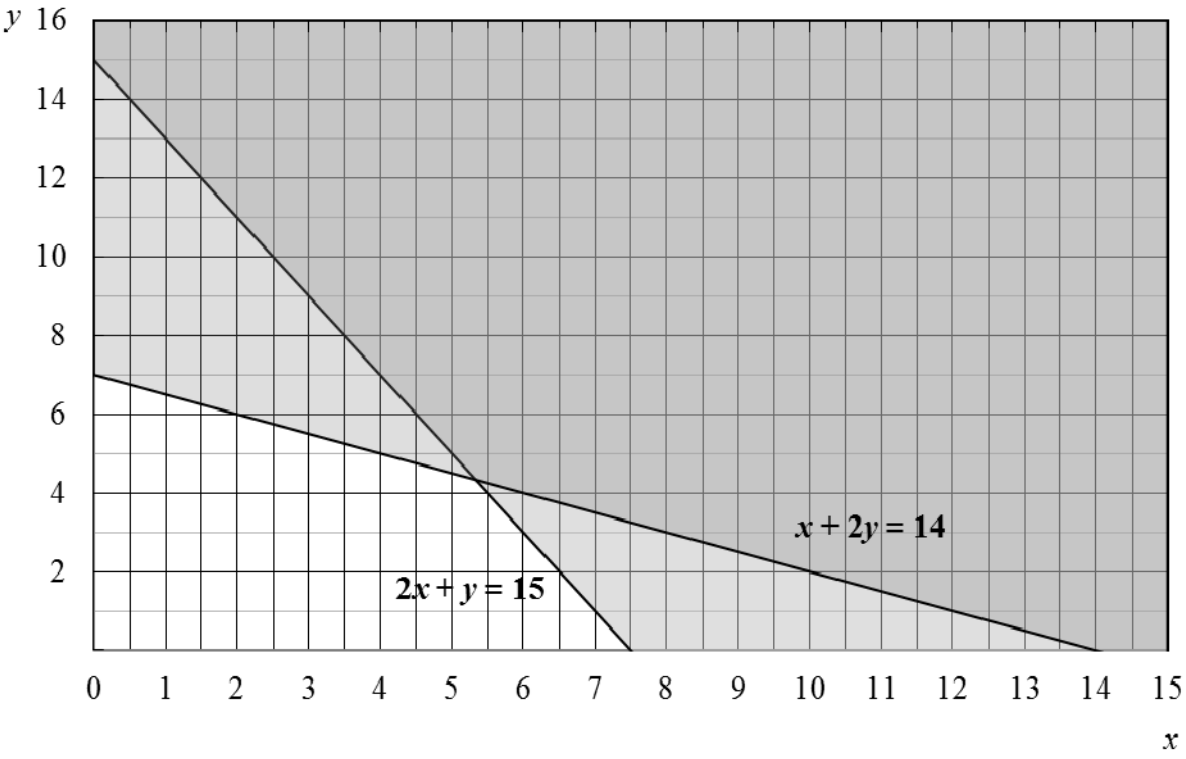
22. The maximum value of $f(x, y) = 2x + y$, subject to the constraints shown in the given figure is

A. 3
B. 7
C. 10
D. 12



NOT TO SCALE

23. The given graph shows the lines $x + 2y = 14$ and $2x + y = 15$ and their respective shaded half planes.



If the feasible solution set is {(14, 0), (0, 15), (5.3, 4.3)}, then the pair of inequalities that represents the feasible region is

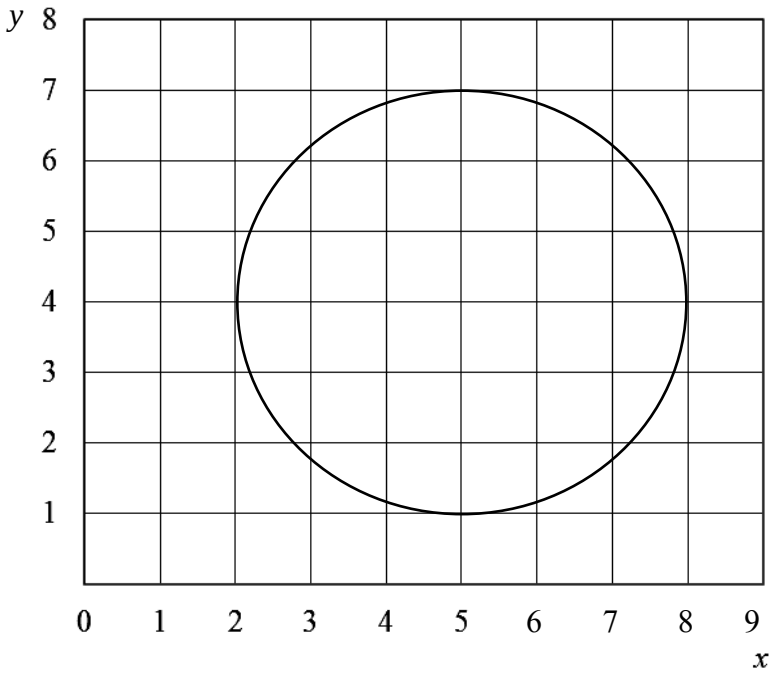
A. $x + 2y \geq 14$
$2x + y \geq 15$

B. $x + 2y \geq 14$
$2x + y \leq 15$

C. $x + 2y \leq 14$
$2x + y \geq 15$

D. $x + 2y \leq 14$
$2x + y \leq 15$

The given figure shows a circle that can be represented by the equation $(x-h)^2+(y-k)^2=r^2$.



Using the given information, answer Q.24 and Q.25.

24. The centre and radius of the circle are

	Centre	Radius
A	(– 5, – 4)	3 units
B	(– 5, – 4)	5 units
C	(5, 4)	3 units
D	(5, 4)	5 units

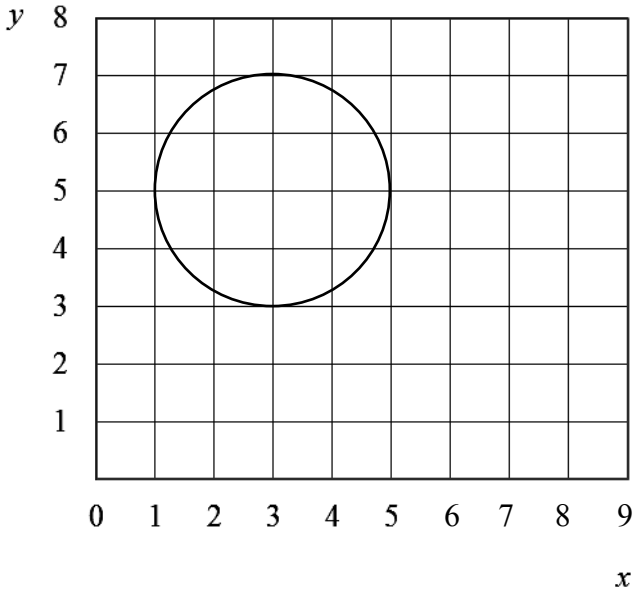
25. The values of *h* and *k* are

	h	***k***
A	– 5	– 4
B	– 4	– 5
C	4	5
D	5	4

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26. The given figure shows a circle that can be presented by the equation

$x^2 + y^2 + 2gx + 2fy + c = 0$.



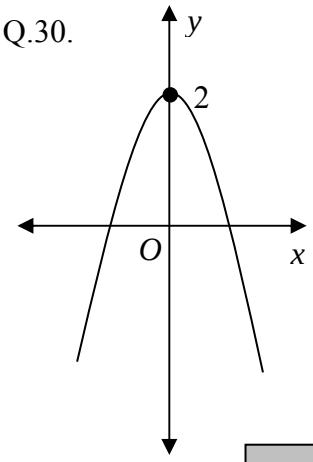
If the value of c is 30, then the equation of circle is

A. $x^2 + y^2 + 6x + 10y + 30 = 0$
B. $x^2 + y^2 + 10x + 6y + 30 = 0$
C. $x^2 + y^2 - 6x - 10y + 30 = 0$
D. $x^2 + y^2 - 10x - 6y + 30 = 0$

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The given parabola can be presented by the equation $(x-h)^2 = -4(y-k)$.

Using the given information, answer Q.27, Q.28, Q.29 and Q.30.



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27. The values of *h* and *k* are

	h	***k***
A	− 2	0
B	0	− 2
C	0	2
D	2	0

28. The focus and vertex are

	Focus	**Vertex**
A	(0, 1)	(0, 2)
B	(0, 2)	(0, 1)
C	(0, − 1)	(0, − 2)
D	(0, − 2)	(0, − 1)

29. The equation of directrix is

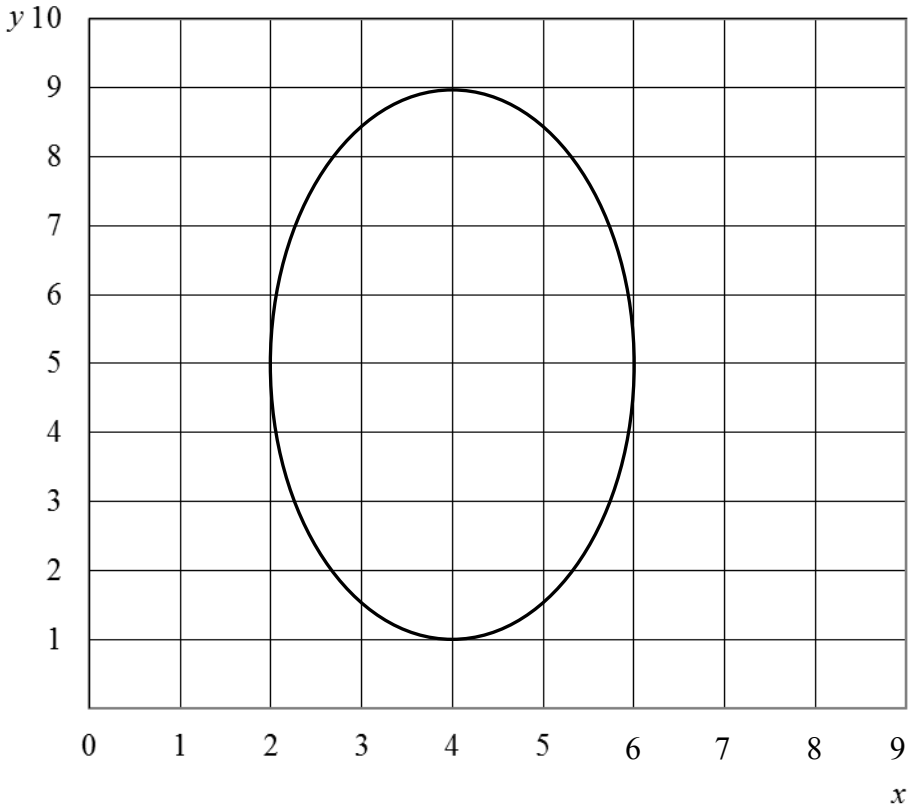
A. $x = 1$
B. $x = 3$
C. $y = 1$
D. $y = 3$

30. The axis of this parabola is

A. $x = 0$
B. $x = 2$
C. $y = 0$
D. $y = 2$

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The ellipse shown in the graph can be represented by the equation $\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1$.



Using the given information, answer Q.31 and Q.32.

31. The values of *h* and *k* are

	h	***k***
A	1	2
B	2	1
C	4	5
D	5	4

32. The values of *a* and *b* are

	a	***b***
A	$\sqrt{2}$	2
B	2	$\sqrt{2}$
C	2	4
D	4	2

33. The equation $x^2 - y^2 - 2x + 2y - 4 = 0$ represents a/ an

A. circle.
B. ellipse.
C. parabola.
D. hyperbola.

34. The equation of the line $x - 3 = 0$, with respect to new parallel axes obtained by shifting the origin to O' (0, – 1), is

A. $X - 3 = 0$

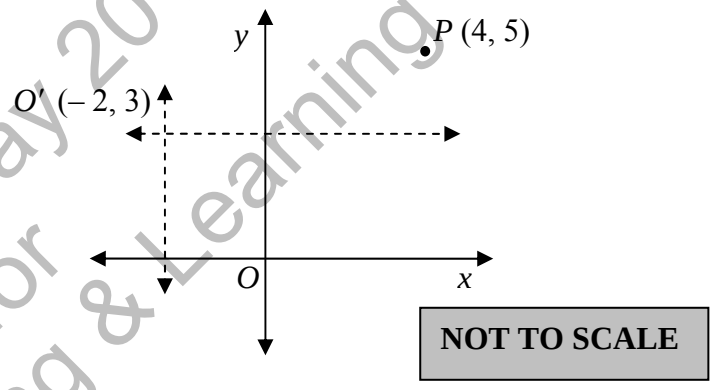
B. $X + 3 = 0$

C. $X + Y - 3 = 0$

D. $X + Y - 4 = 0$

35. With respect to the origin O (0, 0), the coordinates of point P are (4, 5). The coordinates of the point P with respect to the new origin O' (– 2, 3) are

A. (2, 8)
B. (6, 8)
C. (6, 2)
D. (– 6, – 2)



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